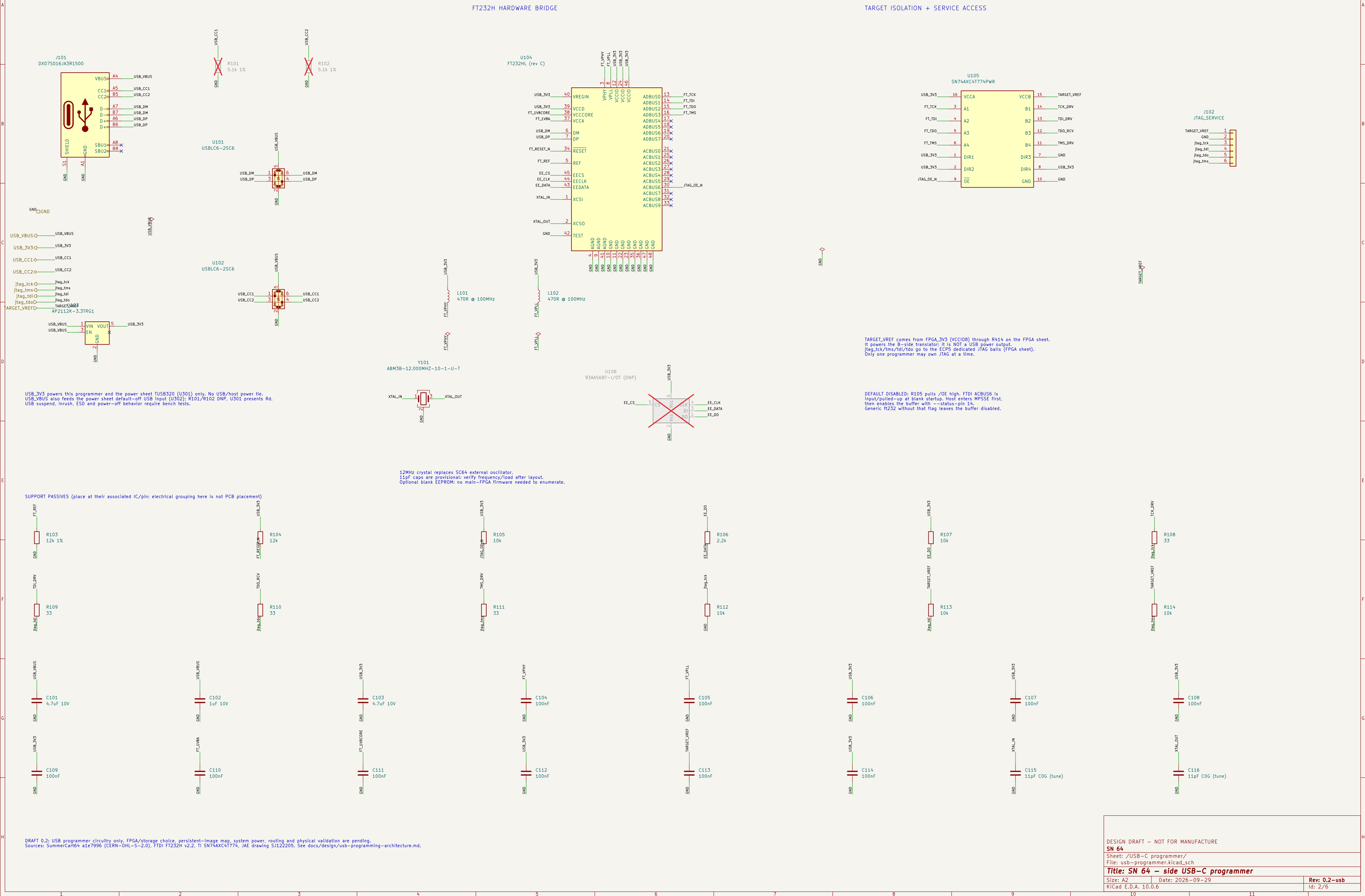


SIDE USB-C: INITIAL PROGRAMMING + RECOVERY

USB 2.0 device: 5V input only. Side location is mandatory: exact PCB/enclosure placement remains open.

FT232H HARDWARE BRIDGE

TARGET ISOLATION + SERVICE ACCESS



DRAFT 0.2: USB programmer circuitry only. FPGA/storage choice, persistent-image map, system power, routing and physical validation are pending.
Sources: SummerCart64 a1e7996 (CERN-DHL-S-2.0), FTDI FT232H v2.2, TI SN74AXC41774, JAE drawing SJ122205. See docs/design/usb-programming-architecture.md.

DESIGN DRAFT – NOT FOR MANUFACTURE

SN 64

Sheet: /USB-C programmer/
File: usb-programmer.kicad_sch

Title: SN 64 – side USB-C programmer

Size: A2 Date: 2026-09-29

KiCad E.D.A. 10.0.6

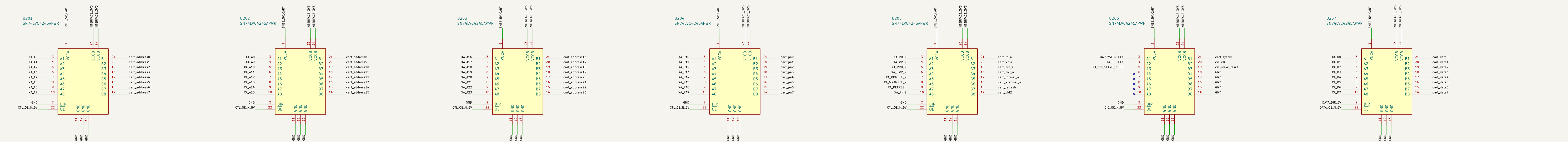
Rev: 0.2-usb

Id: 2/6

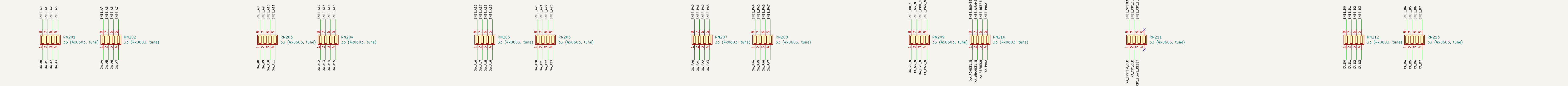
SNES CARTRIDGE –SIDE TRANSLATION AND PROTECTION

SN74LVC4245A: A port = 5 V FPGA (VCCA = SNES_V5_CART), B port = 3.3 V FPGA (VCCB = INTERFACE_3V3); DIR/DE are referenced to VCCA (TI SCAS375K); DIR=L=H=A (FPGA drives socket); DIR=H=A=B; DE=H: Isolation.

U201–U206: console outputs, DIR strapped to GND (fixed FPGA→cartridge), U201–U205 share CTL_DE_N_5V (bridge ctl_de_n); U206 (SYSTEM_CLK + CIC_CLK + CIC_SLAVE_RESET only) has its own CIC_DE_N_5V (sn64_top_snes_cic_de_n, enabled from the sequencer IFACE state).
U207: DD–DT, DIR and DE both FPGA–controlled (bridge data_dir / data_de_s). Unused B inputs tied to GND; unused A outputs no-connect.

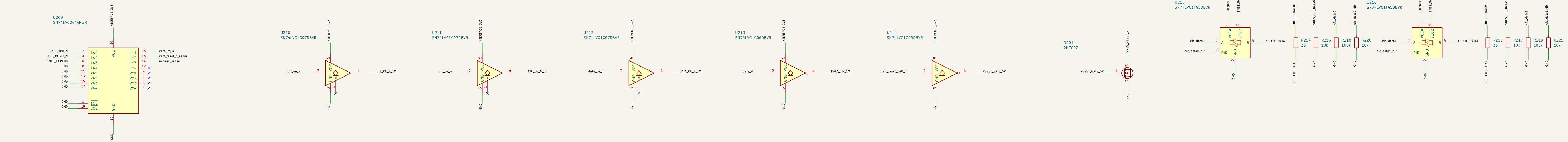


CARTRIDGE OUTPUT DAMPING: 33 R arrays at the translator A pins (source–series). Values are provisional; tune after SI measurement.



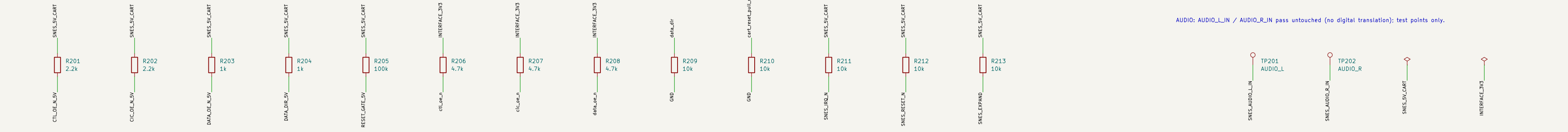
SENSE + RELEASE PATHS

U209 (SN74LVC244A) at 3.3 V, 5.5 V–tolerant inputs with buff. recovers /IRQ, /RESET, EXPAND (CIC data pin via U215/U216).
U210–U212 74LVC1G07 / U213 74LVC1G06, 3.3 V open–drain outputs (5.5 V tolerant, I_{OFF}) pulled to SNES_V5_CART.
No 5 V pull–up reaches an FPGA pin; signal rail off or FPGA unconfigured → every translator /OE released high (isolated).
/RESET: Q201 open–drain sink, RESET_GATE_5V pulled to 5 V; cartridge held in reset unless INTERFACE_3V3 is up AND the FPGA drives cart_reset_pull_n=1.



/OE and DIR edge timing: the bridge releases the data octet one A6.56 ns clock before turnaround. R203/R204 = 1k keep the passive rising edge short (about 10 pF load; tau < 10 ns). R201/R202 2.2k drive fove/one /OE inputs where only permit–loss timing matters. Verify on hardware.

PULL/BIAS: /IRQ, /RESET, EXPAND 10k to cartridge 5 V (OpenSFC keeps pull–ups on IRQ/EXPAND); CIC_DATA0 1 10k pull–downs (R216/R217). Values provisional.



DECOUPLING: one 100 nF per supply pin (place at the pin) plus one 10 uF bulk per rail.



DBAT 0.51–cart: schematic circuit candidate only. No PCB, SI simulation, power sequencing test, ESD part or cartridge test has been performed.
SNES_V5_CART and INTERFACE_3V3 are hierarchical inputs from the power sheet; #U201/#U202 are kept because the power–sheet switch outputs are passive pins (ERC only).
Open: ticket ESD error selection, final damping/pull values (CIC pull–downs value to confirm against a rise by CIC).
Source: TI SCAS375K (SN74LVC4245A), SCAS14AG (SN74LVC244A), SCE5296AG (SN74LVC1G07), SCE2956B (SN74LVC1G06), SCE515N (SN74LVC145).
See docx/design/cart–interface–schematic.mxd.

HIERARCHICAL PORTS – CARTRIDGE SIDE (root J2 net labels)

SNES_A[0..23]→SNES_A[0..23]
SNES_PA[0..7]→SNES_PA[0..7]
SNES_DI[0..7]→SNES_DI[0..7]
SNES_RD_N→SNES_RD_N
SNES_WR_N→SNES_WR_N
SNES_PRO_N→SNES_PRO_N
SNES_PWR_N→SNES_PWR_N
SNES_ROMSEL_N→SNES_ROMSEL_N
SNES_WRAMSEL_N→SNES_WRAMSEL_N
SNES_REFRESH→SNES_REFRESH
SNES_PHI2→SNES_PHI2
SNES_SYSTEM_CLK→SNES_SYSTEM_CLK
SNES_CIC_CLK→SNES_CIC_CLK
SNES_CIC_SLAVE_RESET→SNES_CIC_SLAVE_RESET
SNES_CIC_DATA0→SNES_CIC_DATA0
SNES_CIC_DATA1→SNES_CIC_DATA1
SNES_IRQ_N→SNES_IRQ_N
SNES_RESET_N→SNES_RESET_N
SNES_EXPAND→SNES_EXPAND
SNES_AUDIO_L_IN→SNES_AUDIO_L_IN
SNES_AUDIO_R_IN→SNES_AUDIO_R_IN
GND→GND
SNES_V5_CART→SNES_V5_CART

PENDING PORTS – FPGA SIDE (bridge port names) = DEFAULT–OFF RAILS FROM FUTURE POWER SHEET

cart_address[0..23]→cart_address[0..23]
cart_pa[0..7]→cart_pa[0..7]
cart_data[0..7]→cart_data[0..7]
cart_rd_n→cart_rd_n
cart_wr_n→cart_wr_n
cart_pro_n→cart_pro_n
cart_pwr_n→cart_pwr_n
cart_romsel_n→cart_romsel_n
cart_wramsel_n→cart_wramsel_n
cart_refresh→cart_refresh
cart_phi2→cart_phi2
cart_sysclk→cart_sysclk
cic_clk→cic_clk
cic_slave_reset→cic_slave_reset
cic_data0_dir→cic_data0_dir
cic_data1_dir→cic_data1_dir
ctl_de_n→ctl_de_n
cic_or_n→cic_or_n
data_or_n→data_or_n
data_dir→data_dir
cart_reset_pull_n→cart_reset_pull_n
cart_irq_n→cart_irq_n
cart_reset_n_sense→cart_reset_n_sense
expand_sense→expand_sense
cic_data0→cic_data0
cic_data1→cic_data1
INTERFACE_3V3→INTERFACE_3V3

DESIGN DRAFT – NOT FOR MANUFACTURE

SN 64

Shiren /SNES cartridge interface/
File: cart–interface.kicad_sch

Title: SN 64 – SNES cartridge interface

Size: A1 Date: 2026–09–29

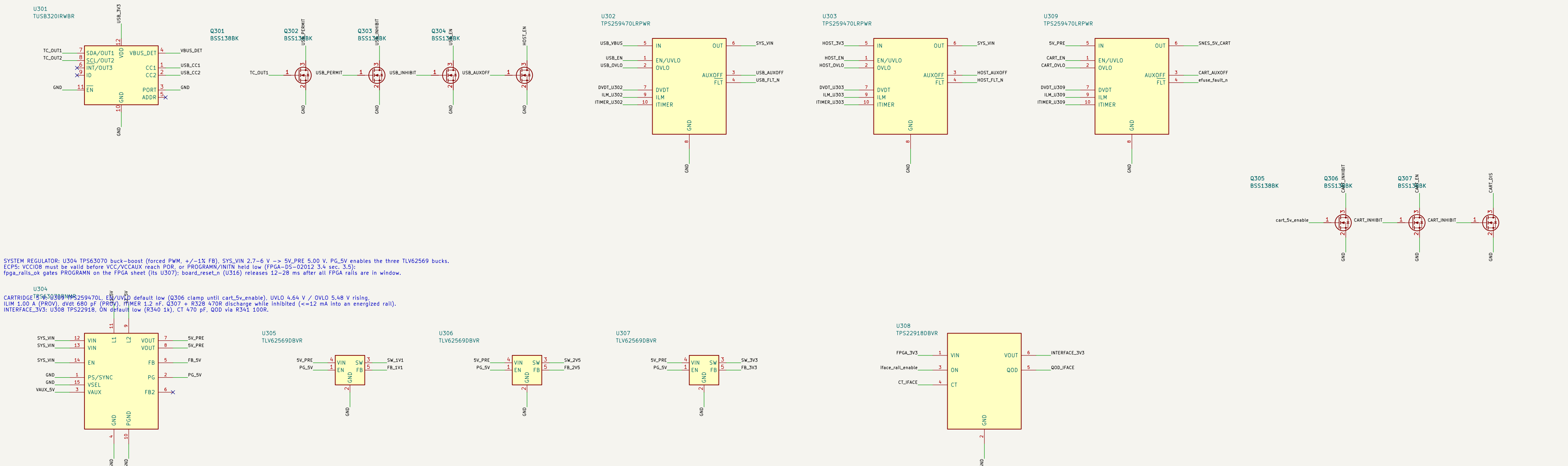
Kicad E.D.A. 10.0.6

Rev: 0.31–cart
16 / 3/6

SN64 POWER – source selection, system regulator, FPGA rails, cartridge eFuse, interface rail, monitors

Implements docs/design/power-architecture.md math and sources in docs/design/power-schematic.md. HOST_12V is not used on this sheet.
Priority power max (1.5VFSVC Figure 8-14): U302 (USB, priority) AUXOFF -> Q304 puis U303 (HOST_3V3, auxillary) EN/VLVO low.
Both eFuses have two reverse-current blocking, no source is ever directly paralleled onto SYS_VIN.

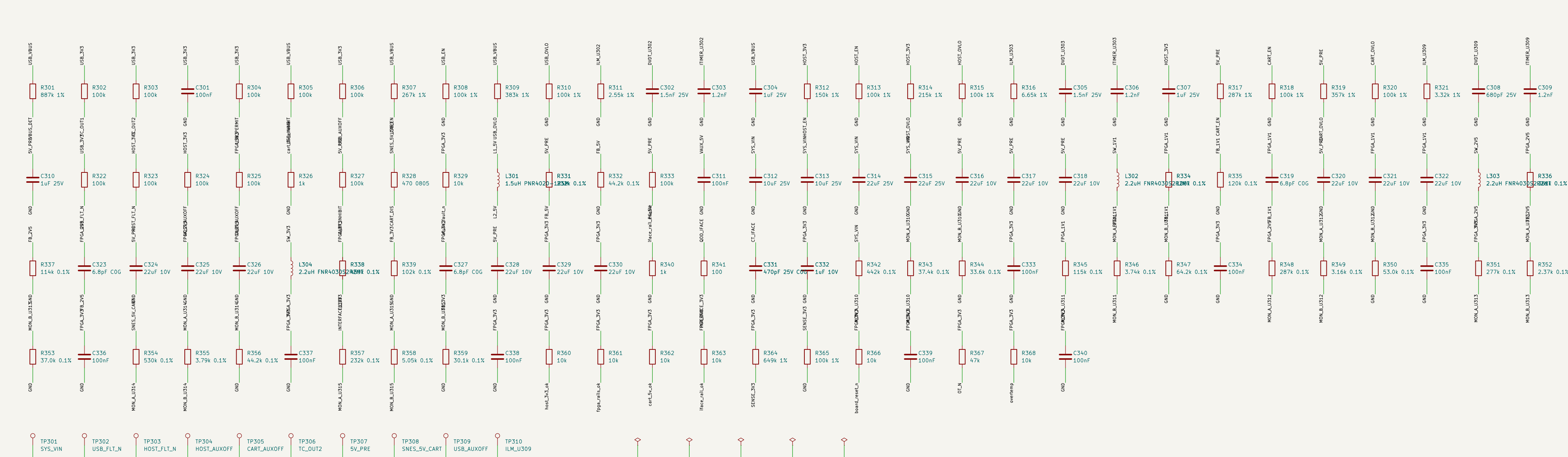
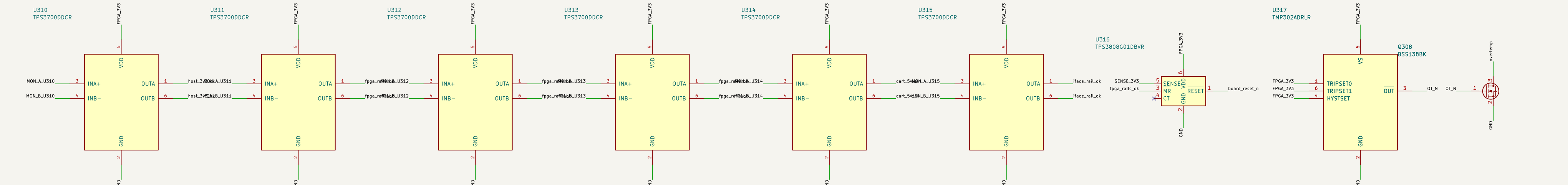
USB permission (default off). U302 enables only when USB8320 OUT1 is low (1.5 A or 3 A advertised) AND USB_3V3 is present.
USB8320: PORT=GNP (OFF), ADDR open (GPIO mode). EN_N=GNP, VDD=USB_3V3 (power register PWR=LO6). The USB sheet 5.1k Rd must be DNP.



SYSTEM REGULATOR: U304 TPS83070 buck-boost (forced PWM, +/-1k F8). SYS_VIN 2.7-6 V -> SV_PFE 5.00 V. P_C3V enables the three TLV62569 bucks.
ECP5: VCC0B must be valid before VCC/VCCALX reach PDR, or PROGRAM/NRST held low (FPGA-D5-R2012 3.4 sec. 3.5).
FPGA_rail_en gates PROGRAMN on the FPGA sheet (U307: board_level= (U316) releases 12-26 ms after all FPGA rails are in window.

U304 CARTRIDGE TPS83070 buck-boost (forced PWM, +/-1k F8). SYS_VIN 2.7-6 V -> SV_PFE 5.00 V. P_C3V enables the three TLV62569 bucks.
ECP5: VCC0B must be valid before VCC/VCCALX reach PDR, or PROGRAM/NRST held low (FPGA-D5-R2012 3.4 sec. 3.5).
FPGA_rail_en gates PROGRAMN on the FPGA sheet (U307: board_level= (U316) releases 12-26 ms after all FPGA rails are in window.

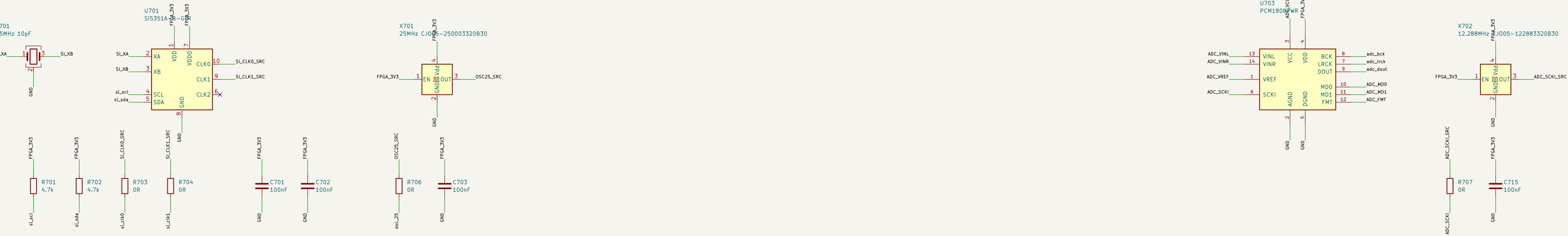
MONITORS (all powered from FPGA_3V3, outputs pulled up to FPGA_3V3 = 3.3 V-safe; supply loss reads NOT OK):
U310 SYS_VIN -> hst_3v3_ok | U311-U313 FPGA rails -> fpga_rail_ok | U314 SNES_SV_CART (reset line) -> cart_5v_ok |
U315 INTERFACE_3V3 -> iface_rail_en | U309 FS (Lck to FPGA_3V3) -> fpga_rail_en | U307 TMP300A + Q306 -> overtemp (1 = hot or sensor dead).



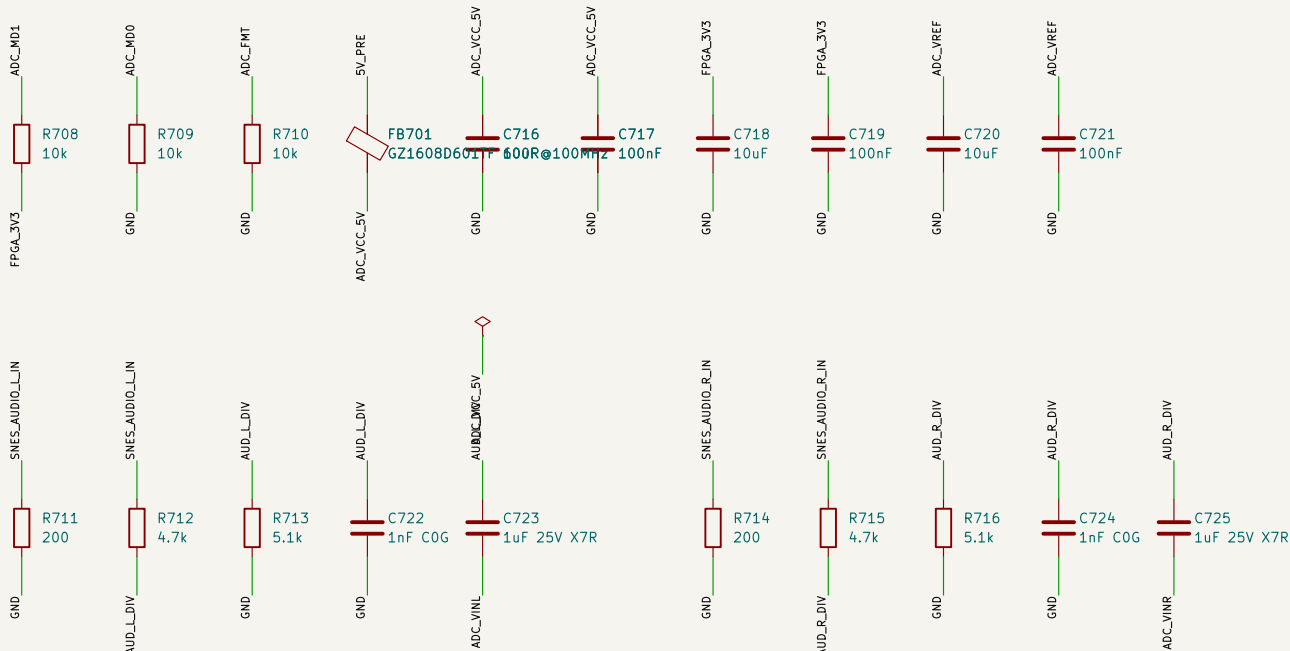
SN 64 – CLOCKS AND CARTRIDGE AUDIO ADC (DRAFT 0.2–av)

CLOCKS (docs/design/clock–plan.md). U701 SI5351A–B–GT (Skyworks SI5351–B Rev 1.3): CLK0 = NTSC master 21.4772727 MHz; CLK1 = PAL master 21.28137 MHz; CLK2 powered down and disabled (register 3 = 0xFC), pin 6 open: programmed at power–on by fpga/tu/sn64_clock_init.sv over si_scl/si_sda (I2C address 0x60, 400 kHz), VDD and VDD0 both FPGA_3V3 (sec. 7.2; VDD0 with or before VDD). One 100 nF per supply pin (sec. 7.1). Y701 25 MHz crystal, CL 10 pF: matches register 183 XTAL_CL = 11b (0x02) in sn64_clock_init.sv (AN619 Rev 0.8). Internal load capacitance only; no external load caps (sec. 7.4). Crystal (YXC Y5K3215L datasheet): ESR <= 50 R, drive level up to 200 uW, C0 <= 3 pF; SI5351 Table 8 needs ESR <= 150 R, max drive level >= 100 uW, CL 6–12 pF. R701/R702 4.7k pull–ups to VDD (Figure 5; >= 1k per Table 20). R703/R704 0R; SI5351 drives 50 R (Table 5, ZO) into a 50 R trace; sec. 7.6 shows an optional 0 R series resistor (EMI). X701 25 MHz 3.3 V CMOS oscillator (FPGA housekeeping, osc_25): pin 1 enabled tied high (run), R706 0R optional damping. Footprint pads differ slightly from the CJ005 suggested layout; PROVISIONAL.

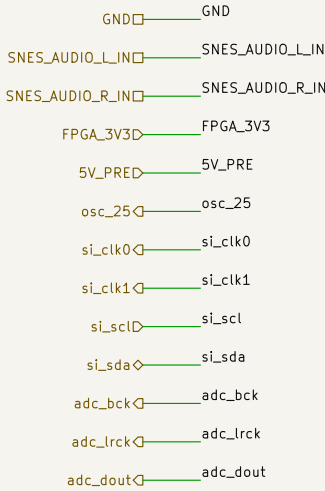
CARTRIDGE AUDIO (SNES socket pins 31 AUDIO_L_IN / 62 AUDIO_R_IN). U703 PCM1808 (TI SLES177B) in I2S MASTER mode, 384 fs; MD1 = H (R708 to VDD), MD0 = L, FMT = L (I2S 24–bit). X702 12.288 MHz = 384 x 32 kHz -> fs = 32,000 kHz (own oscillator; asynchronous to the SNES master; the FPGA must rate–match). BCK = 64 fs = 2,048 MHz, outputs at VDD = FPGA_3V3. VCC (analog 5 V) = 5V_PRE through FB701 + 10 uF + 100 nF; VDD (digital) = FPGA_3V3; AGND/DGND to GND at the device. Input per channel: R711/R714 200 R to GND, = console load (OpenSFC SHVC–CPU–01 R95/R96 200 R; its 1 uF + 10 k summing input in parallel). Divider 4.7k / (5.1k || 60k ADC input) = 0.5900; 5.0 Vp–p at the pin (largest swing from the 5 V–only cartridge supply) -> 2.5 Vp–p; ADC full scale 0.6 x VCC = 3.0 Vp–p (2.85 Vp–p at 4.75 V). Input impedance at the pin 200 R || 9.4k = 196 R (console: 200 R || 10k = 196 R), 1 nF COG; 67.7 kHz anti–alias pole (internal filter –3 dB at 1.3 MHz), 1 uF coupling; 2.6 Hz high–pass. Reference level: sd2snes Rev F (CS4344 at 3.3 V, 0.65 x VA typ = 2.15 Vp–p, 100 R + 470 R) gives –0.56 Vp–p into the 200 R console load -> –20.6 dBFS here. SNES audio does NOT go to N64_AUDIO_L/R (no requirement; see doc).



VIDEO AND AUDIO TO THE CONSOLE (docs/design/console–video–path.md): the SNES picture and sound reach the N64/M64 over the cartridge bus (N64 endpoint frame window) and are shown on the console's own output (N64 AV jack, M64 HDMI). This board has no video output of its own. The HDMI section of draft 0.1–av (J701 type–A, U702 TPD125016, 22 nF TMDS coupling, SI5351 CLK2 pixel clock) was removed in 0.2–av.



HIERARCHICAL PORTS (shared net–name contract: GND/SNES_AUDIO_* are root labels, the rest pending on the FPGA/power sheets)



DRAFT 0.2–av: schematic circuit candidate only. No PCB, SI simulation or audio measurement has been performed. FPGA_3V3 and 5V_PRE are hierarchical inputs from the power sheet (ERC origins there: #FLG304 and the TP563070 VOUT). Provisional: 0R series values, 200 R console–equivalent load (OpenSFC value, verify on SHVC–CPU–01), X7R 1 uF coupling, footprints of X701/X702. Sources and SHA–256 of every datasheet: libraries/av–provenance.json. See docs/design/av–clock–schematic.md.